Tlotlo Oepeng

Data Scientist & Machine Learning Engineer

Ramotswa, Botswana · +267 73 221 763 · 03peng@gmail.com
linkedin.com/in/Oepeng · github.com/Soot-bit

Profile

Data scientist with an MSc in AI (Distinction, Google DeepMind Scholar) and a BSc in Physics (Dean's Prize). I apply statistical analysis, data mining and machine learning to large structured and unstructured data sets, from images and documents to tabular and simulation data, and turn the results into models, dashboards and clear recommendations. Currently building production machine learning and data pipelines at De Beers Group, with a software engineering approach: version control, CI/CD, data integrity checks and reproducible evaluation.

Core Strengths

Data Analysis	Statistical analysis, data mining, pattern discovery, regression, classification, clustering, anomaly detection, forecasting, causal inference
Unstructured Data	Images (segmentation and detection), documents (retrieval and question answering), simulation and sensor data
Data Pipelines	Python data processing, SQL, training and deployment pipelines on Azure, CI/CD, data integrity and validation checks
Visualisation & BI	Plotly and Matplotlib dashboards, Excel (intermediate), stakeholder reporting, familiarity with Power BI and Databricks
Toolkits	Python, Pandas, NumPy, SciPy, scikit-learn, PyTorch, Jupyter, SQL, Bash, C/C++
Software Engineering	Git, FastAPI, Docker, REST APIs, ONNX, modular and tested code

Professional Experience

ML Engineer, De Beers Group Jan 2026 to Present
Hybrid, UK remote

- Build and evaluate models on large volumes of industrial image data for a production diamond-detection system, delivering analysis and stakeholder-ready updates on model quality and trade-offs.
- Built a human-in-the-loop data labelling platform by extending the Label Studio codebase: the model pre-annotates images, experts correct them, and uncertainty-based sampling selects which data to label next.
- Maintain CI/CD pipelines that move data and models from training on Azure GPU compute to deployment on production machines (ONNX).
- Build APIs and a retrieval-augmented assistant that lets engineers query unstructured machine manuals and documentation.

Research Collaboration, Florida State University Feb 2025 to Present
With Prof. Harrison Prosper, part-time

- Develop analytics workflows and ML models on physics simulation data, separating data generation, training, diagnostics and reporting so results can be checked and reproduced.
- Awarded a USD 1,200 research excellence prize for this collaboration.

ML Research Scientist, African Institute for Mathematical Sciences Aug 2024 to Feb 2025
South Africa

- Supervised graduate students on machine learning projects, and reviewed and debugged their Python code for model evaluation, optimisation and numerical reasoning.
- Ran parallel training workflows on cloud services and HPC infrastructure (GCP, CUDA GPUs, TPUs).

Freelance Data Scientist, SearchQualify 2021 (6 months)
Remote

- Delivered general data science projects sourced through a remote AI and data talent platform: data analysis, modelling and evaluation, including deep learning methods.

IT Technician, Olasef Printing Service Oct 2019 to Feb 2021
Botswana

- Automated recurring service tasks and resolved hardware and software issues across daily operations.

Education

MSc Mathematical Sciences (AI), University of Cape Town (AIMS South Africa) 2023 to 2024
Dissertation with Distinction, Google DeepMind Scholar. Coursework: Causality, Bayesian Modelling, Simulation-Based Inference

BSc Physics, University of Botswana 2015 to 2019
Dean's Prize, highest graduating student in the Faculty of Science

Selected Projects

GemAI: Diamond Price Model GitHub
TabNet, PyTorch, Pandas, FastAPI, Docker
- Built a structured-data model for diamond price estimation, served through a FastAPI endpoint with Docker, with repeatable evaluation and consistent scoring for downstream pricing decisions.

Movie Recommender GitHub
C++, OpenMP, PyBind, NumPy, Pandas
- Built a collaborative-filtering system on 25 million user ratings. Found power-law behaviour in user activity and used it to design a sparse data structure, which with OpenMP parallelism cut training and inference time by more than 50x.

TrackFormer (MSc dissertation) GitHub
PyTorch Lightning, Transformers, Scientific Python
- Built a transformer model for particle track fitting on TrackML, a $\sim$5TB realistic detector data set with noise hits. Found that quantile loss is far more robust than MSE on noisy, high-volume data.

Publications

- T. Oepeng and H. B. Prosper, *Physics-informed neural networks for photon orbits in the Schwarzschild metric*, Machine Learning and the Physical Sciences (ML4PS) Workshop, NeurIPS 2026 (accepted).
- C. B. Tabi, E. Wamba, T. Oepeng and T. C. Kofané, *Dissipative quantum fluctuations-induced droplets in Bose-Einstein mixtures*, Physics Letters A 564, 131088 (2025).

Additional Information

- Languages: English (C1), Setswana (native). Transcripts available on request.

References

Thabiso Jacobs Science and Technology Implementation Manager, De Beers Group
thabiso.jacob@debeersgroup.com

Prof. Harrison Prosper Kirby W. Kemper Endowed Professor, Florida State University
hprosper@fsu.edu · Research collaboration supervisor